Year 12 Maths - Vectors

Similar questions - Position vectors in geometric figures

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $ABCD$ is a quadrilateral. The position vectors of A and B are $2\mathbf{i}+\mathbf{j}$ and $6\mathbf{i}+4\mathbf{j}$ respectively.

(a) Find the length AB. **[1]**

(b) The position vector of C is $p\mathbf{i}+(p-1)\mathbf{j}$, where $p>2$. Given that $BC=AB$, find the position vector of C. **[4]**

(c) Given that $ABCD$ is a parallelogram, find the position vector of D. **[2]**

(d) Name the quadrilateral $ABCD$, giving a reason, and show that its diagonals have the same midpoint. **[3]**

2 Points A and B have position vectors $\begin{pmatrix}-2\\3\end{pmatrix}$ and $\begin{pmatrix}1\\1\end{pmatrix}$ respectively.

(a) Point C has position vector $\begin{pmatrix}p\\-3\end{pmatrix}$, and A, B and C lie on a straight line. Find p. **[3]**

(b) Point D has position vector $\begin{pmatrix}q\\2\end{pmatrix}$ and angle $ABD=90^\circ$. Find q. **[3]**

Question 1

(a) $\overrightarrow{AB} = \binom{4}{3}$, so $AB = 5$

(b) $BC^2 = (p-6)^2 + (p-5)^2 = 25$, so $2p^2 - 22p + 36 = 0$, i.e. $p^2 - 11p + 18 = 0$ and $(p-2)(p-9) = 0$. As $p > 2$, $p = 9$ and $C = 9\mathbf{i} + 8\mathbf{j}$

(c) $\mathbf{d} = \mathbf{a} + \mathbf{c} - \mathbf{b} = \binom{2+9-6}{1+8-4} = \binom{5}{5}$, so $D = 5\mathbf{i} + 5\mathbf{j}$

(d) A rhombus: a parallelogram with two adjacent sides equal ($AB = BC = 5$). Midpoint of $AC = (5.5, 4.5)$ and of $BD = (5.5, 4.5)$

Question 2

(a) $\overrightarrow{AB} = \binom{3}{-2}$, $\overrightarrow{BC} = \binom{p-1}{-4}$. Parallel: $\dfrac{p-1}{3} = \dfrac{-4}{-2} = 2$, so $p = 7$

(b) $\overrightarrow{BA} = \binom{-3}{2}$, $\overrightarrow{BD} = \binom{q-1}{1}$. Perpendicular: $-3(q-1) + 2 = 0$, so $q = \frac{5}{3}$